Lecture 12, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

Method of Successive Approximations

Ex.: $\begin{cases} y' = 2x(1+y) \\ y(0) = 0 \end{cases}$

As an integral equation: $y(x) = \int_0^x 2s(1+y(s))\,\mathrm{d}s$

Define the sequence of functions: $y_n(x) = \int_0^x 2s(1+y_{n-1}(s))\,\mathrm{d}s$

Start with $y_0(x) = 0$.

$y_1(x) = \int_0^x 2s(1+y_0(s))\,\mathrm{d}s = \int_0^x 2s(1+0)\,\mathrm{d}s = x^2$

$$y_2(x) = \int_0^x 2s(1 + y_1(s))\,\mathrm{d}s = \int_0^x 2s(1 + s^2)\,\mathrm{d}s = x^2 + \frac{x^4}{2}$$

$$y_3(x) = \int_0^x 2s(1 + y_2(s))\,\mathrm{d}s = \int_0^x 2s(1 + s^2 + \frac{s^4}{2})\,\mathrm{d}s = x^2 + \frac{x^4}{2} + \frac{x^6}{6}$$

Guess:

$$y_n(x) = x^2 + \frac{(x^2)^2}{2!} + \frac{(x^2)^3}{3!} + \cdots + \frac{(x^2)^n}{n!} = \sum_{k=1}^{n} \frac{(x^2)^k}{k!}$$

$$
\begin{aligned}
y_{n+1}(x) &= \int_0^x 2s\left(1+\sum_{k=1}^{n}\frac{(s^2)^k}{k!}\right)\mathrm{d}s \\
&= \int_0^x \left(2s+\sum_{k=1}^{n}\frac{(2s^{2k+1}}{k!}\right)\mathrm{d}s \\
&= x^2+\sum_{k=1}^{n}\frac{2(x^2)^{2k+2}}{(2k+2)k!} \\
&= x^2+\sum_{k=1}^{n}\frac{(x^2)^{k+1}}{(k+1)k!} = x^2+\sum_{k=1}^{n}\frac{(x^2)^{k+1}}{(k+1)!} \\
&= x^2+\sum_{k=2}^{n+1}\frac{(x^2)^k}{k!} = \sum_{k=1}^{n+1}\frac{(x^2)^k}{k!}
\end{aligned}
$$

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Rewrite:

$$y_n = \sum_{k=1}^{n} \frac{(x^2)^k}{k!} + 1 - 1$$

$$= \sum_{k=0}^{n} \frac{(x^2)^k}{k!} - 1$$

$y_n \to_{n\to\infty} \boxed{e^{x^2} - 1 = y(x)}$

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Second Order Linear Equations

Ex: $x^2\,y'' + \sin(x)\,y' + 5\,y = 3 + x$

Rewrite in standard form: $y'' + \frac{\sin(x)}{x^2}\,y' + \frac{5}{x^2}\,y = \frac{3+x}{x^2}$

This is equation is inhomogeneous. The associated homogeneous equation is $y'' + \frac{\sin(x)}{x^2}\,y' + \frac{5}{x^2}\,y = 0$

Principle of Superposition for Linear Homogeneous Equations

Ex: $y'' - 3y' + 2y = 0$ (linear homogeneous equation)

$y_1(x) = e^x$ and $y_2(x) = e^{2x}$ are solutions of the equation. Any linear combination $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is also a solution of the same equation (c_1 and c_2 can be any number; they're free parameters)

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Ex: $y'' - y = 4$ (linear inhomogeneous)

$y_1(x) = -4$ and $y_2(x) = \mathrm{e}^x - 4$ are solutions of the equation, but

the sum $y_1(x) + y_2(x)$ is not.

Ex: $y'' + 2\,y\,y' = 0$ (nonlinear)

$y_1(x) = 1$ and $y_2(x) = \frac{1}{x}$ are solutions of the equation, but the sum

$y_1(x) + y_2(x)$ is not.

The principle of superposition means that any linear combination of solutions of a linear homogeneous equation is also a solution of the equation.

This is in general not true for inhomogeneous or nonlinear equations.

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A generic 2nd order linear homogeneous equation:

$$y'' + p(x)\,y' + q(x)\,y = 0$$

If $y_1(x)$ and $y_2(x)$ are solutions it means that:

$$y_1'' + p(x)\,y_1' + q(x)\,y_1 = 0 \text{ and } y_2'' + p(x)\,y_2' + q(x)\,y_2 = 0$$

Verify that $c_1\,y_1 + c_2\,y_2$ is also a solution:

$$c_1\,y_1'' + c_2\,y_2'' + p\,c_1\,y_1' + p\,c_2\,y_2' + q\,c_1\,y_1 + q\,c_2\,y_2 = 0$$

$$c_1\,[y_1'' + p\,y_1' + q\,y_1] + c_2\,[y_2'' + p\,y_2' + q\,y_2] = 0$$

no matter what c_1 and c_2 are. This is the

principle of superposition.

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2nd Order Linear IVP

A second order initial value problem will have up to two initial conditions. Ex:

$$\begin{cases} y'' - 3y' + 2y = 0 \\ y(0) = 1 \\ y'(0) = 2 \end{cases}$$

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Solve this and similar I.V.P. in three steps:

1) Find 2 solutions of the equation, $y_1(x)$ and $y_2(x)$.

2) Combine the solutions linearly: $y(x) = c_1\, y_1(x) + c_2\, y_2(x)$

(general solution).

3) Apply the initial conditions to $y(x)$ to determine c_1 and c_2 and

get the solution to the IVP.

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For this example:

1) $y_1(x) = \mathrm{e}^x$ and $y_2(x) = \mathrm{e}^{2x}$

2) $y(x) = c_1\,\mathrm{e}^x + c_2\,\mathrm{e}^{2x}$ (general solution to the equation) and

$y'(x) = c_1\,\mathrm{e}^x + 2\,c_2\,\mathrm{e}^{2x}$

3) $y(0) = c_1 + c_2 = 1$ and $y'(0) = c_1 + 2\,c_2 = 2$ which has solutions

$c_1 = 0$ and $c_2 = 1$. So the solution to the I.V.P. is

$y(x) = 0 * \mathrm{e}^x + 1 * \mathrm{e}^{2x} = \mathrm{e}^{2x}$

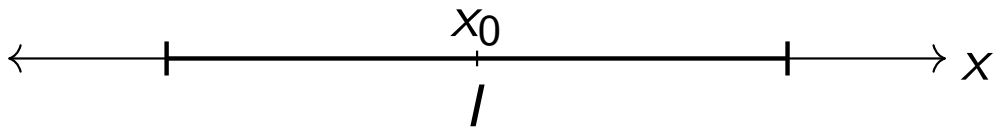
Theorem of Existence and Uniqueness for 2nd Order Linear IVP

$$\begin{cases} y'' + p\,y' + q\,y = f(x) \\ y(x_0) = \alpha \\ y'(x_0) = \beta \end{cases}$$

If $p(x), q(x)$, and $f(x)$ are continuous in an interval I that includes

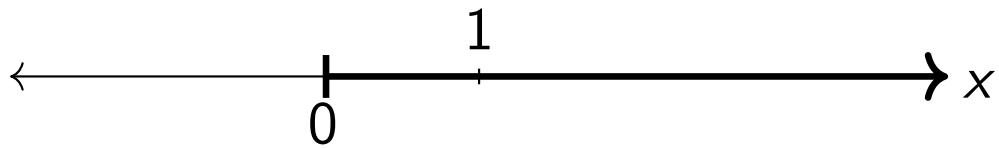
x_0 then $\exists$ one and only one solution to the IVP over all of I

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$$\begin{cases} y'' + \frac{1}{x}y' - \frac{9}{x^2}y = 0 \\ y(1) = 0 \\ y'(1) = 1 \end{cases}$$



$I = (0, +\infty)$

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